dl.py - C:/Users/pings/Desktop/dl.py (3.14.2)

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```
#Exp No:2 Upload and Analyze the data set given in csv format and perform data
#preprocessing and visualization.
#Description: Use sample data set sales-data.csv.

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)
# Display the first few rows of the DataFrame
print(df.head())
# Check for missing values
print(df.isnull().sum())
# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
# Summary statistics
print(df.describe())
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
'Sales': 'sum',
'Quantity': 'sum'
}).reset_index()
print(product_summary)
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('SalesOver Time')
plt.show()
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
```

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plt.show()
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)
# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

Screenshot saved

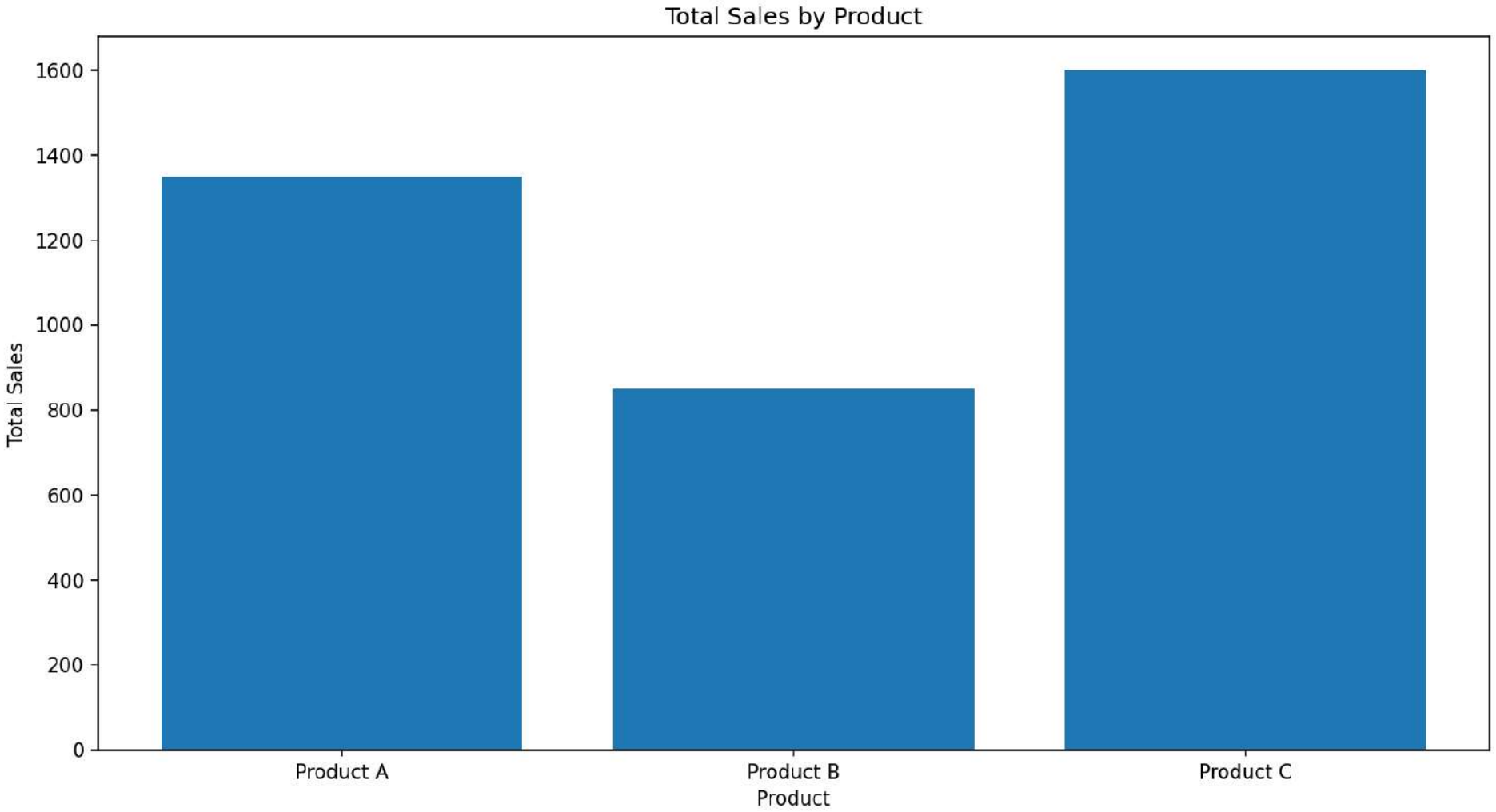
IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/pings/Desktop/dl.py ==========
         Date    Product  Sales  Quantity Region
0  2023-01-01  Product A    200         4  North
1  2023-01-02  Product B    150         3  South
2  2023-01-03  Product A    220         5  North
3  2023-01-04  Product C    300         6   East
4  2023-01-05  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
                      Date      Sales   Quantity
count                   16   16.00000  16.000000
mean   2023-01-08 12:00:00  237.50000   5.375000
min    2023-01-01 00:00:00  150.00000   3.000000
25%    2023-01-04 18:00:00  187.50000   4.000000
50%    2023-01-08 12:00:00  225.00000   5.500000
75%    2023-01-12 06:00:00  302.50000   7.000000
max    2023-01-16 00:00:00  340.00000   8.000000
std                    NaN   64.44636   1.746425
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```

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Figure 1


Total Sales by Product
Total Sales
1600
1400
1200
1000
800
600
400
200
0
Product A
Product B
Product C
Product

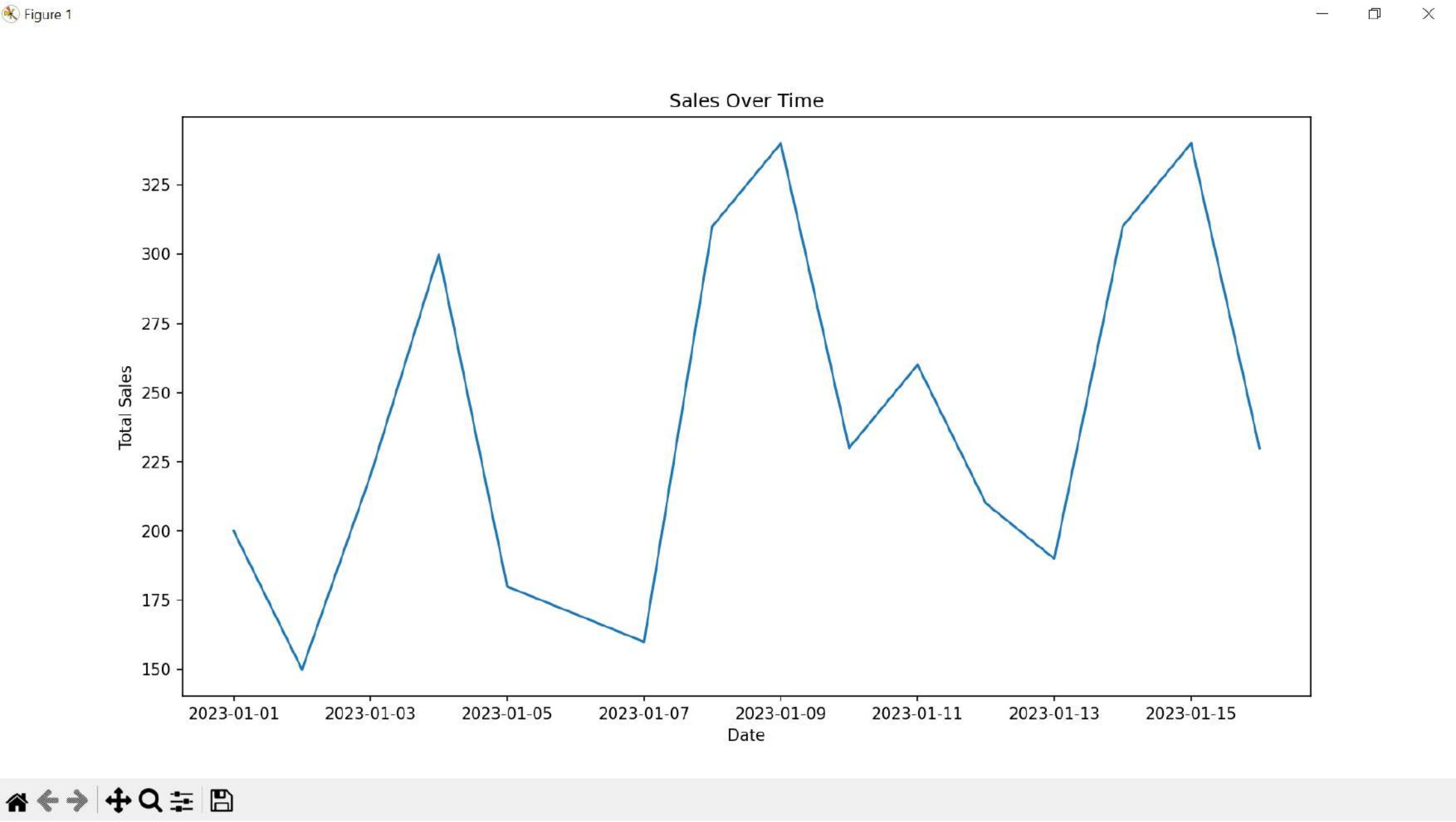
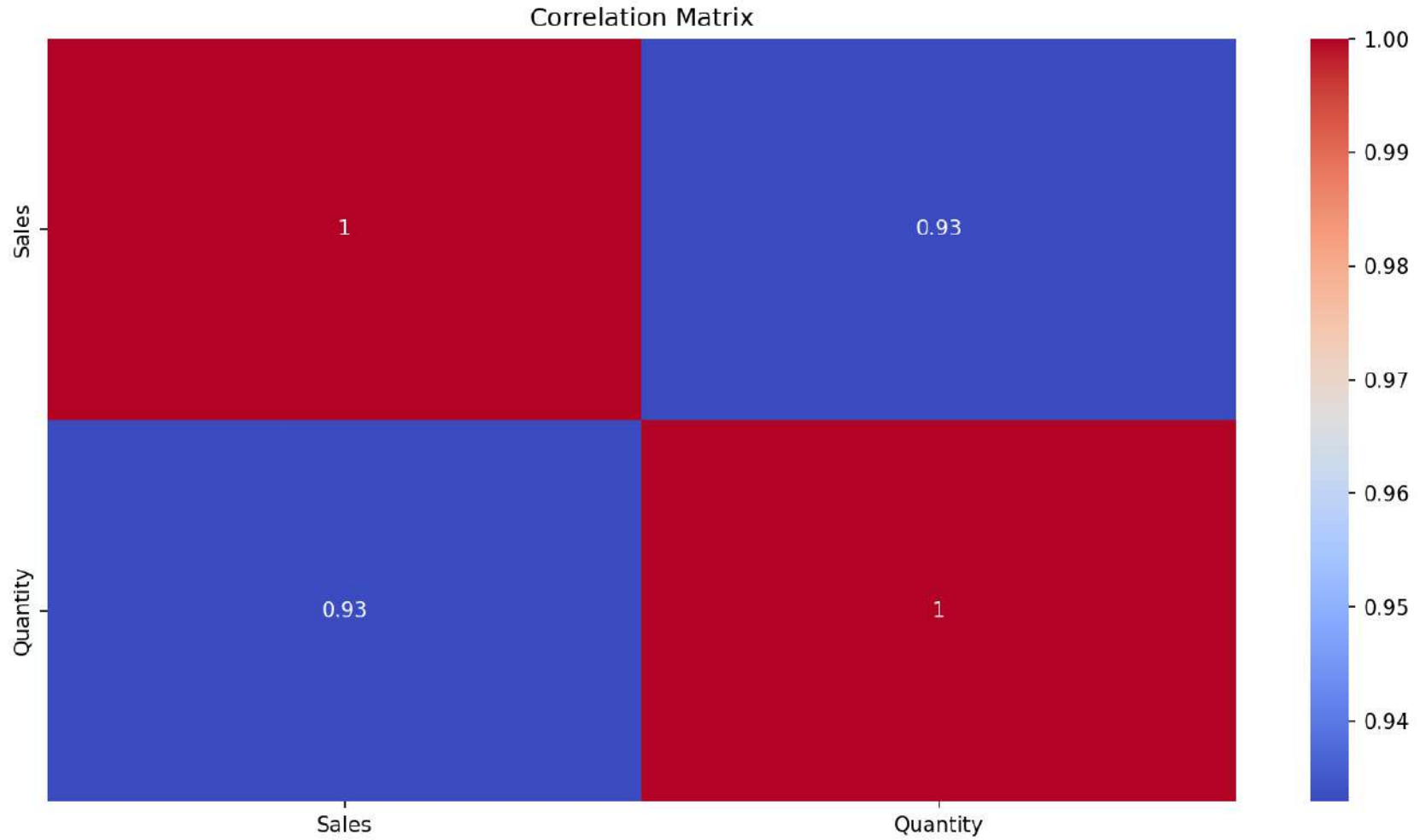

Figure 1
Sales Over Time
Total Sales
325
300
275
250
225
200
175
150
2023-01-01
2023-01-03
2023-01-05
2023-01-07
2023-01-09
2023-01-11
2023-01-13
2023-01-15
Date

Figure 1


Correlation Matrix
Sales
Quantity
1
0.93
0.93
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95
0.94